

10907 Pattern Recognition

Lecturers

Prof. Dr. Ilija Bogunovic (ilija.bogunovic@unibas.ch)

Tutors

Alexandra Spitzer (alexandra.spitzer@unibas.ch)

Felicitas Haag (felicitas.haag@unibas.ch)

Tommaso Antonini (t.antonini@unibas.ch)

Rahel Kempf (rahel.kempf@unibas.ch)

Sanjoth Jagannath Rai (sanjoth.jagannathrai@unibas.ch)

Introductory Problem Set

This is the first weekly problem set. Its purpose is to outline the mathematical and programming background needed to navigate the course confidently. There are no weekly hand-in assignments; instead, a problem set will be posted every other week and discussed in the exercise sessions. You are strongly encouraged to engage deeply with the material, as it is a prerequisite for developing the skills required to pass the exams. Remember that each exercise will not appear on the exam in exactly the same form, but it serves to outline the relevant area of knowledge. Each exercise has a difficulty level from $\star$ to $\star\star\star$.

1 Math

1 Linear Algebra

Exercise 1 (System of Linear Equations $\star$)

Consider the following system of linear equations,

$$\begin{aligned} 2x - 7y - 9z &= k_1 \\ 3x + 3y + 4z &= k_2 \\ 11x + 2y + 3z &= k_3 \end{aligned}$$

What are the values of k_1, k_2, k_3 , for which the system above has solutions? If we set $k_1 = k_2 = k_3 = 0$, what are the possible values of x, y and z ?

Exercise 2 (Elementary Row and Column Operations $\star$)

Let M be a block matrix given as

$$M = \begin{bmatrix} I_n & \mathbf{0} \\ \mathbf{0} & AB \end{bmatrix},$$

where $A \in \mathbb{R}^{s \times n}$, $B \in \mathbb{R}^{n \times t}$ and I_n is an $n \times n$ identity matrix. Use elementary row and column operations to transform this matrix into

$$N = \begin{bmatrix} B & I_n \\ \mathbf{0} & A \end{bmatrix}.$$

Provide an explanation of the row and column operations you used.

Exercise 3 (Rank of a Matrix $\star$)

Let M and N be as in the previous exercise, $M = \begin{bmatrix} I_n & \mathbf{0} \\ \mathbf{0} & AB \end{bmatrix}$ and $N = \begin{bmatrix} B & I_n \\ \mathbf{0} & A \end{bmatrix}$, with dimensions as before.



1. What conclusions can you draw regarding the rank of matrix $\mathbf{N}$ from the ranks of matrices $\mathbf{A}$ and $\mathbf{B}$?
2. Using the results from part 1 of the exercise and the previous exercise, prove the inequality:
 $\text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}) - \text{rank}(\mathbf{AB}) \leq n$.
3. If $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{AB}) = s$, what is the rank of matrix $\mathbf{B}$?

Exercise 4 (Eigenvalues and Positive Definiteness)

Let $\mathbf{A} \in \mathbb{R}^{s \times s}$ be an $s \times s$ real matrix such that $\mathbf{x}^T \mathbf{A} \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^s$. Show that $\mathbf{A}$ does not have any negative eigenvalues.

2 Probability

Exercise 5 (System of Linear Equations with Random Variables ★★)

For the system of linear equations given below:

$$\begin{aligned} 2X - 7Y + 9Z &= K_1 \\ 3X + 3Y + 4Z &= K_2 \\ 5X + 2Y + 5Z &= K_3 \end{aligned}$$

The random variables X, Y, Z are independent and normally distributed such that $X \sim \mathcal{N}(0, 1)$, $Y \sim \mathcal{N}(1, 4)$ and $Z \sim \mathcal{N}(0, 9)$.

- What are the distributions of K_1, K_2 and K_3 ?
- What is the joint probability density function (pdf) of K_1, K_2 and K_3 , $p(k_1, k_2, k_3)$?

Recall that a normally distributed random variable X with mean μ and variance σ^2 (written $X \sim \mathcal{N}(\mu, \sigma^2)$) has the probability density function (pdf)

$$p_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

so that for some a, b such that $a \leq b$ we have

$$\mathbb{P}[a \leq X \leq b] = \int_a^b p_X(x) dx.$$

Exercise 6 (Sum of Bernoulli Distribution ★★★)

Alice and Bob decide to play a game. They toss a biased coin n times. If it comes up heads at least $n/2$ times, Bob wins; otherwise Alice wins. The total number of heads is

$$S = \sum_{i=1}^n X_i$$

where X_i is an independent Bernoulli random variable with mean p , that is,

$$\mathbb{P}(X_i = 1) = p, \quad \mathbb{P}(X_i = 0) = 1 - p.$$

Assume that $p = 0.52$.

1. What is the law of the random variable S ? Give its name, its probability mass function (p.m.f.) and its mean.

2. Bob only wants to play if he has a 90% chance of winning.

For a given n , what is the expected number of heads?

Compute the probability that Bob wins when $n = 2$.

Can you figure out for which values of n Bob will participate in the game?

Hint: One way to do this is by using Hoeffding's inequality: let Z_i be independent and identically distributed random variables such that $0 \leq Z_i \leq 1$, then

$$\mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n Z_i - \mathbb{E}(Z_i) \geq t\right) \leq \exp(-nt^2).$$

3. Alice and Bob continue the game for a long, long time. What can you say about the value or distribution of S as $n \rightarrow \infty$? What about the value or distribution of $\frac{S}{n}$? And what about the value or distribution of $\frac{S - np}{\sqrt{n}}$?

Exercise 7 (Expectations ★★)

The expectation of a non-negative random variable X with probability density function $f_X(x)$ is

$$\mathbb{E}(X) = \int_0^\infty x f_X(x) dx.$$

Show that an alternative way to compute it is by the following useful identity:

$$\mathbb{E}(X) = \int_0^\infty \mathbb{P}(X > t) dt.$$

Exercise 8 (From Derivatives to Gradients ★)

- Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 3x^2 - 4x + 1$. Compute $f'(x)$. Solve $f'(x) = 0$. Is this a minimizer or maximizer of f ?
- Recall that the derivative is a limit of slopes, $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$. For $g(x) = x^2$ and $x = 1$, evaluate the quotient $\frac{g(1+h) - g(1)}{h}$ exactly (by hand, symbolically in h) and then for $h = 0.1$ and $h = 0.01$. Compare with $g'(1)$. What is the error as a function of h ?
- Now let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1^2 + 3x_1x_2 + 2x_2^2$. Compute the two *partial* derivatives $\frac{\partial f}{\partial x_1}$ and $\frac{\partial f}{\partial x_2}$. To compute $\frac{\partial f}{\partial x_1}$, treat x_2 as a constant and differentiate in x_1 exactly as in part 1, and vice versa.
- Collect the partial derivatives into the *gradient*

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(\mathbf{x}) \\ \vdots \\ \frac{\partial f}{\partial x_d}(\mathbf{x}) \end{bmatrix} \in \mathbb{R}^d.$$

Write $\nabla f(\mathbf{x})$ for the f of part 3 and evaluate it at $\mathbf{x} = (1, 1)^T$. *Sanity check to internalise now and use forever:* f maps $\mathbb{R}^d \rightarrow \mathbb{R}$, so $\nabla f(\mathbf{x})$ is a vector in $\mathbb{R}^d$ — the same shape as the input, never a scalar and never a matrix.

5. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$,

$$f(x_1, x_2) = \exp(2x_1 - 3x_2).$$

Using the chain rule, compute $\frac{\partial f}{\partial x_1}$ and $\frac{\partial f}{\partial x_2}$, and collect them into $\nabla f(\mathbf{x})$. Evaluate the gradient at $\mathbf{x} = (0, 0)^T$.



6. Now consider a more deeply nested function,

$$f(x_1, x_2) = \exp(x_1^2 + 3x_1x_2 + 2x_2^2).$$

Define

$$q(x_1, x_2) = x_1^2 + 3x_1x_2 + 2x_2^2, \quad f(x_1, x_2) = \exp(q(x_1, x_2)).$$

First compute $\nabla q(\mathbf{x})$. Then use the multivariate chain rule to show that

$$\nabla f(\mathbf{x}) = \exp(q(\mathbf{x})) \nabla q(\mathbf{x}).$$

Write out both components explicitly and evaluate the gradient at $\mathbf{x} = (1, 0)^T$.

7. Let $\mathbf{w}, \mathbf{x} \in \mathbb{R}^d$ and $y \in \mathbb{R}$, and consider the loss

$$L(\mathbf{w}) = \frac{1}{2} (\exp(\mathbf{w}^T \mathbf{x}) - y)^2.$$

This function is a composition of several operations. Introduce the intermediate quantities

$$z = \mathbf{w}^T \mathbf{x}, \quad \hat{y} = \exp(z), \quad L = \frac{1}{2} (\hat{y} - y)^2.$$

Compute

$$\frac{\partial L}{\partial \hat{y}}, \quad \frac{\partial \hat{y}}{\partial z}, \quad \nabla_{\mathbf{w}} z,$$

and combine them using the chain rule to obtain $\nabla_{\mathbf{w}} L(\mathbf{w})$. Check that the resulting gradient has the same shape as $\mathbf{w}$. Finally, evaluate it for

$$\mathbf{w} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad y = 3.$$

2 Coding

To get your hands dirty with Python, you are asked to write some basic array operations related to linear algebra and probability. Specifically, you need to complete the blank functions in `sample.py`.

Exercise 9 (Basic Computation ★)

1. Write a function `check_span` that takes as the inputs a vector $\mathbf{y}$ and a matrix $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_n]$, where $\mathbf{a}_i$'s are the column vectors of the matrix. The function outputs a Boolean variable. The output is `True` if $\mathbf{y}$ is in the span of column vectors in $\mathbf{A}$. Otherwise, the output is `False`.
2. Write a function `estimate_mean_and_covariance` to compute the mean and the unbiased covariance matrix of a set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$. The set of vectors are passed to the function as a matrix $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n]$. The unbiased covariance matrix is defined as:

$$\mathbf{Q} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$$

where $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$.